

Recombining Knowledge after Patent Eligibility Tightening

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Motivation: Innovation Adjustment to Changing Economic Incentives

The innovation environment is inherently dynamic:

- technological opportunities, market conditions, and appropriability can change over time

Changes in economic incentives redirect innovative activity:

- A natural response is to reallocate innovative effort toward activities with higher relative returns

This paper:

- Knowledge recombination as a margin of innovation adjustment

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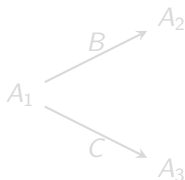
Existing knowledge can be reused in subsequent innovation:

- knowledge generated by prior innovation can serve as an input into new innovation

$$A_1 \longrightarrow A_2$$

The same knowledge can be reused in different ways:

- the same knowledge can be combined with different complementary knowledge
- innovators can choose among alternative knowledge combinations



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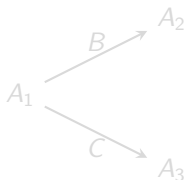
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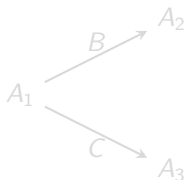
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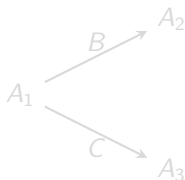
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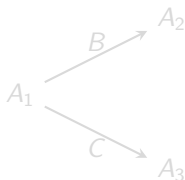
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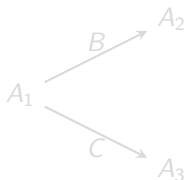
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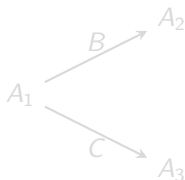
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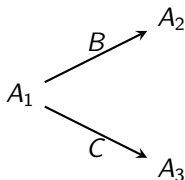
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Alice: A Shock to the Value of Existing Knowledge

The Alice Decision (2014):

- a U.S. Supreme Court decision on a patent infringement lawsuit ◀ Exogeneity
- tightened patent eligibility for abstract ideas
- disproportionately affected software and business methods (Chien and Wu, 2018)

Why Alice?

- Patent grants in highly exposed technologies recovered after the initial disruption ◀ Trend
⇒ Innovation recovered within affected technological domains
- Inventions combining multiple technology classes are less likely to receive Alice-related rejections ◀ Proportions
⇒ Cross-technology recombination provides a potential margin of adjustment

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Main Findings

- **Innovators respond to changing incentives through knowledge recombination**
 - ▶ innovations in more exposed technologies combine a broader set of technology classes after Alice
 - ▶ recombination shifts innovation toward less-exposed technological combinations
- **Recombination opportunities shape the extent of adjustment**
 - ▶ the recombination response is stronger when affected technologies have greater opportunities to combine with other technologies
- **Firm technological boundaries shape access to these opportunities**
 - ▶ firms with greater access to recombination opportunities sustain more innovation in affected technologies

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Alice Exposure

1. Technology-class exposure (a knowledge component): Exposure

$$Exposure_i = Pr(\text{Alice-related rejection} \mid \text{technology class } i)$$

- Estimated using applications **filed before** but **examined after** Alice
 - ▶ technological content was chosen before the decision
 - ▶ examination outcomes reflect the post-Alice eligibility standard

2. Patent exposure (an innovation):

$$Exposure_p = \sum_{i \in p} w_{ip} Exposure_i$$

- weighted exposure across technology classes combined in patent p Weights

3. Firm exposure (an innovator):

$$Exposure_f = \sum_p Exposure_{fp}$$

- $Exposure_{fp}$: the exposure of patent p owned by firm f

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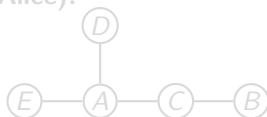
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Recombination Opportunities in the Technology Space

How easily can a focal technology be recombined with other technologies?

⇒ It depends on the local structure of the technology space

Technology Network (pre-Alice):



More and stronger links to other technologies

⇒ Greater recombination opportunities

Measuring Recombination Opportunities:

1. **Relatedness between technology classes:** $R_{ij} = \frac{N_{ij}}{\sqrt{N_i N_j}}$

where N_i is the number of historical patents containing class i , and N_{ij} is the number containing both i and j

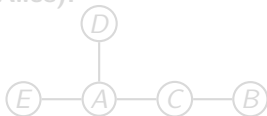
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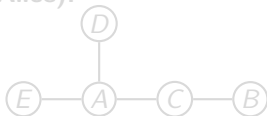
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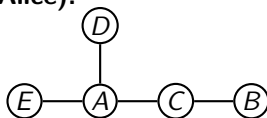
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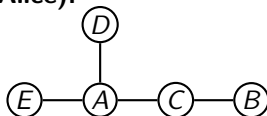
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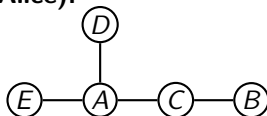
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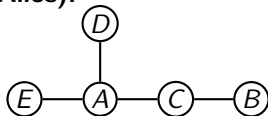
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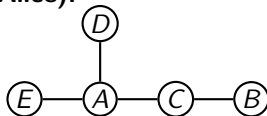
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Firm Boundaries and Recombination Capacity

Can innovators access the complementary knowledge needed for recombination?

⇒ It depends on the firm's existing technological knowledge

Firm f 's Knowledge Portfolio K_f (pre-Alice):



Measuring Firm Recombination Capacity:

1. Accessible recombination opportunities for firm f in technology i :

$$RO_{fi} = \sum_{j \neq i} R_{ij} \mathbb{I}(j \in K_f)$$

2. Firm-level recombination capacity:

$$RC_f = \sum_i w_{fi} RO_{fi}$$

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Firm f 's Knowledge Portfolio K_f (pre-Alice):



Measuring Firm Recombination Capacity:

1. Accessible recombination opportunities for firm f in technology i :

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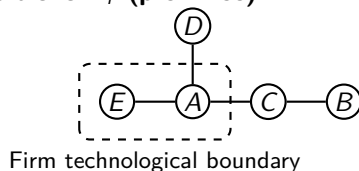
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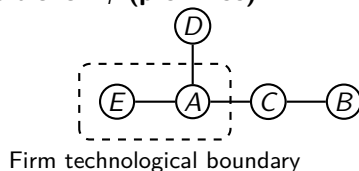
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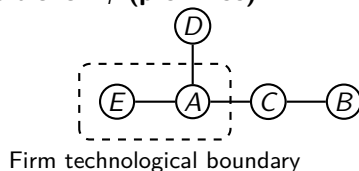
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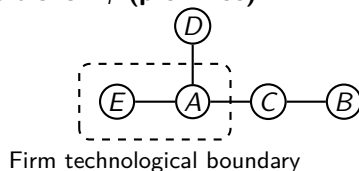
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Cross-Technology Recombination after Alice

After Alice, inventions in more exposed technologies recombine across more technology classes and become less exposed overall.

- **Sample:** US patent applications filed in 2006–2023
- **Treatment:** Alice exposure of the primary technology class

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Continuous Treatment:

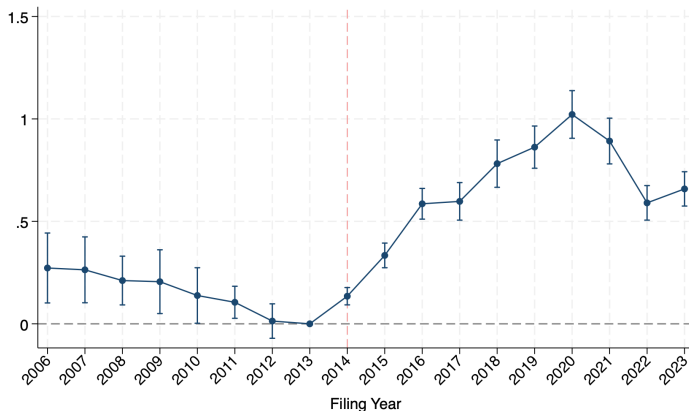
	Num. of Classes		Exposure Reduction	
	(1)	(2)	(3)	(4)
Class Exposure $\times$ Post	0.523*** (7.30)	0.530*** (5.92)	0.191*** (9.82)	0.193*** (7.36)
Primary Class FE	Yes	Yes	Yes	Yes
Filing Year FE	Yes	Yes	Yes	Yes
Clustering	Primary Class	Primary Class	Primary Class	Primary Class
Pseudo/Adj. R^2	0.0325	0.0327	0.4477	0.5112
Num. of Observations	6586138	4772492	6586138	4772492
Sample	Pub Apps	Grants	Pub Apps	Grants
Model	PPML	PPML	OLS	OLS

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Dynamic Effect on Num. of Classes:

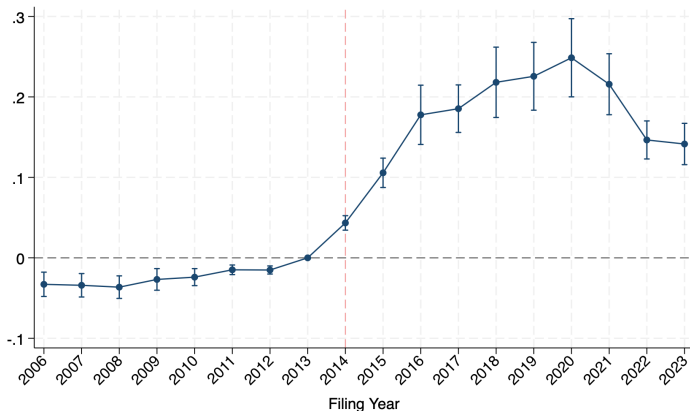


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Discrete Version:

	Num. of Classes		Exposure Reduction	
	(1)	(2)	(3)	(4)
Med. Exposure $\times$ Post	0.0570*** (2.85)	0.0589*** (3.27)	0.000521** (2.04)	0.000540** (2.53)
High Exposure $\times$ Post	0.137*** (3.36)	0.138*** (3.33)	0.0416*** (2.65)	0.0397** (2.40)
Primary Class FE	Yes	Yes	Yes	Yes
Filing Year FE	Yes	Yes	Yes	Yes
Clustering	Primary Class	Primary Class	Primary Class	Primary Class
Pseudo/Adj. R^2	0.0326	0.0328	0.4321	0.4968
Num. of Observations	6586138	4772492	6586138	4772492
Sample	Pub Apps	Grants	Pub Apps	Grants
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Recombination Opportunities Shape Adaptive Responses

The recombination response to Alice is stronger when affected technologies have greater recombination opportunities.

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Heterogeneous Effect by Recombination Opportunities:

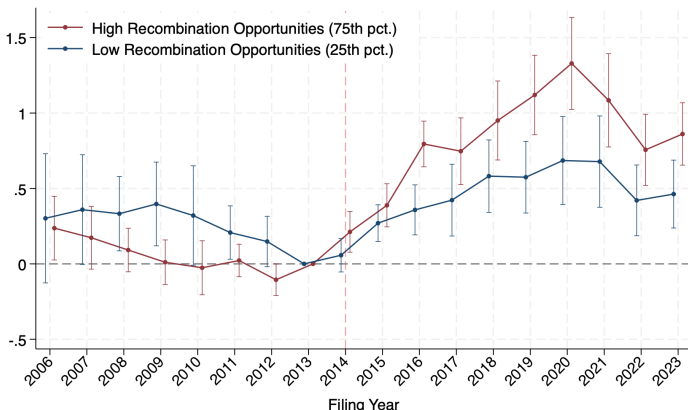
	Num. of Classes		Exposure Reduction	
	(1)	(2)	(3)	(4)
Class Exposure $\times$ Post	0.542*** (9.59)	0.545*** (7.22)	0.197*** (24.36)	0.199*** (18.10)
Recombination Opportunities $\times$ Post	-0.0269*** (-2.74)	-0.0283*** (-2.75)	-0.000512* (-1.67)	-0.000675** (-2.02)
Class Exposure $\times$ Recombination Opportunities $\times$ Post	0.454** (2.33)	0.521*** (2.62)	0.0719*** (2.88)	0.0827*** (3.27)
Primary Class FE	Yes	Yes	Yes	Yes
Filing Year FE	Yes	Yes	Yes	Yes
Clustering	Primary Class	Primary Class	Primary Class	Primary Class
Pseudo/Adj. R^2	0.0326	0.0328	0.4479	0.5114
Num. of Observations	6586138	4772492	6586138	4772492
Sample	Pub Apps	Grants	Pub Apps	Grants
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Dynamic Effect: High vs. Low Recombination Opportunities:

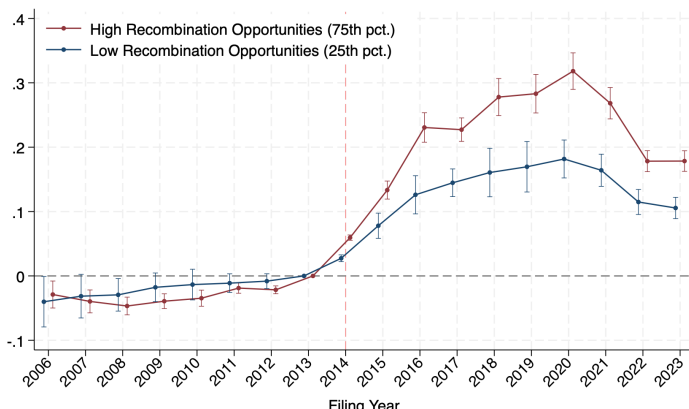


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Event Study (Exposure Reduction):



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- **Treatment:** Alice exposure of the primary technology class, heterogeneous by recombination opportunities

Discrete Version:

	Num. of Classes		Exposure Reduction	
	(1)	(2)	(3)	(4)
Med. Exposure $\times$ Post	0.0560*** (3.12)	0.0602*** (3.61)	0.000554** (2.26)	0.000561*** (2.62)
High Exposure $\times$ Post	0.157*** (4.48)	0.162*** (4.68)	0.0545*** (4.81)	0.0540*** (4.52)
Recombination Opportunities $\times$ Post	-0.0273*** (-3.31)	-0.0294*** (-3.29)	0.0000354** (2.24)	0.0000412*** (2.59)
Med. Exposure $\times$ Recombination Opportunities $\times$ Post	0.0167 (0.70)	0.0152 (0.79)	-0.000209 (-1.02)	-0.000151 (-0.85)
High Exposure $\times$ Recombination Opportunities $\times$ Post	0.0939** (2.36)	0.0981** (2.40)	0.0369*** (3.37)	0.0369*** (3.36)
Primary Class FE	Yes	Yes	Yes	Yes
Filing Year FE	Yes	Yes	Yes	Yes
Clustering	Primary Class	Primary Class	Primary Class	Primary Class
Pseudo/Adj. R^2	0.0326	0.0328	0.4321	0.5142
Num. of Observations	6586138	4772492	6586138	4772492
Sample	Pub Apps	Grants	Pub Apps	Grants
Model	PPML	PPML	OLS	OLS

Recombination Capacity to Sustain Patenting

Firms with greater recombination capacity are better able to sustain patenting in Alice-exposed technologies.

- **Sample:** Public firms with pre-Alice patent portfolios
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Patenting across Technology Exposure:

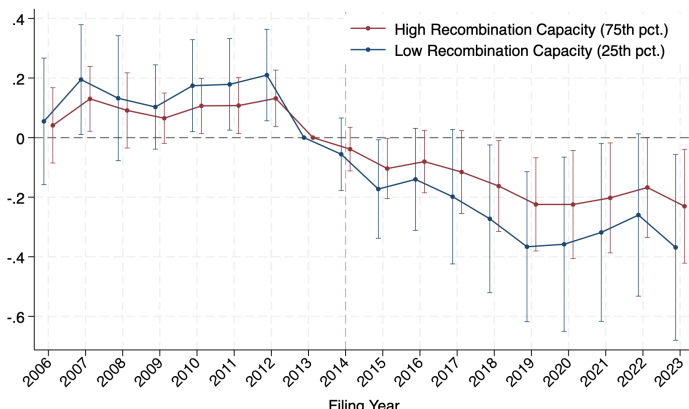
	Number of Patent Applications					
	Total		High-Exposure Tech		Low-Exposure Tech	
	(1)	(2)	(3)	(4)	(5)	(6)
Firm Exposure $\times$ Post	-0.0184** (-2.30)	-0.0908 (-1.43)	-0.0283*** (-3.79)	-0.290*** (-3.34)	-0.00141 (-0.13)	-0.0616 (-0.67)
Recombination Capacity $\times$ Post		0.0558 (1.02)		0.0923 (1.18)		0.0300 (0.50)
Firm Exposure $\times$ Recombination Capacity $\times$ Post		0.0244 (1.14)		0.0888*** (3.15)		0.0216 (0.69)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm
Pseudo R^2	0.9636	0.9636	0.8853	0.8868	0.9568	0.9569
Num. of Observations	29998	29998	16234	16234	26365	26365

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Dynamic Effect on Patenting in High-Exposure Technologies:



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Technology-Side Opportunities Alone Are Not Enough:

	Number of Patent Applications		
	Total	High-Exposure Tech	Low-Exposure Tech
	(1)	(2)	(3)
Firm Exposure $\times$ Post	-0.0184** (-2.41)	-0.0239** (-2.24)	0.000414 (0.05)
Recombination Potential $\times$ Post	0.0850 (0.83)	-0.00951 (-0.07)	0.0431 (0.41)
Firm Exposure $\times$ Recombination Potential $\times$ Post	-0.0239 (-0.80)	-0.0386 (-0.97)	0.0166 (0.41)
Firm Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes
Clustering	Firm	Firm	Firm
Pseudo R^2	0.9636	0.8854	0.9569
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Recombination Capacity and Firm Performance

More exposed firms experience larger declines in performance, but these losses are smaller among firms with greater recombination capacity.

- **Sample:** Public firms with pre-Alice patent portfolios
- **Treatment:** Pre-Alice firm exposure to Alice

	log(Sales)		Sales/Employees		Log(MVE)	
	(1)	(2)	(3)	(4)	(5)	(6)
Firm Exposure $\times$ Post	-0.0102 (-1.14)	-0.0853** (-2.56)	-0.949 (-0.38)	-41.02** (-2.04)	-0.0171 (-1.32)	-0.160*** (-3.53)
Recombination Capacity $\times$ Post		0.0161 (1.00)		34.81*** (2.98)		0.0630*** (3.22)
Firm Exposure $\times$ Recombination Capacity $\times$ Post		0.0268** (2.36)		12.23* (1.90)		0.0486*** (3.06)
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm
Adj. R^2	0.9606	0.9606	0.6955	0.6958	0.9285	0.9286
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Additional Evidence on Knowledge Recombination and Adaptation

Recombination extends beyond technology classifications:

- More exposed technologies draw on more diverse inventor teams 
- More exposed technologies draw on more diverse knowledge through backward citations 

Alternative knowledge components: individual patents

- Exposed patents are increasingly cited by less-exposed inventions after Alice 
- Continuation-in-part filings recombine existing inventions with new knowledge and result in lower Alice exposure 

Firm adaptation through existing technological knowledge:

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- Adaptation occurs primarily through reuse of technologies already within firm boundaries 

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The scope for recombination depends on technological opportunities:

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Firm technological boundaries determine access to these opportunities:

- firms with greater access sustain more innovation in affected technologies and experience smaller performance losses

Takeaway: Changes in economic incentives can redirect innovation not only by changing *where* innovators invest, but also by changing *how existing knowledge is combined*.

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Takeaway: Changes in economic incentives can redirect innovation not only by changing *where* innovators invest, but also by changing *how existing knowledge is combined*.

Conclusion

Knowledge recombination is a margin of innovation adjustment:

- after Alice, innovations in more exposed technologies combine a broader set of knowledge components and shift toward less-exposed combinations

The scope for recombination depends on technological opportunities:

- the recombination response is stronger when affected technologies have greater opportunities to combine with other technologies

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Thanks!

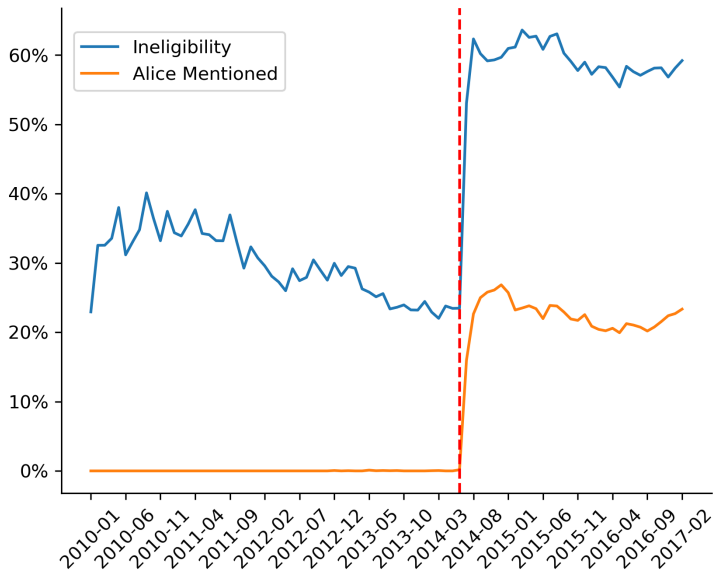
email: danfeng.yue@hotmail.com

website: danfengy.github.io

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Immediate Increase in Patent Examination Rejections (High-Exposure Technologies)



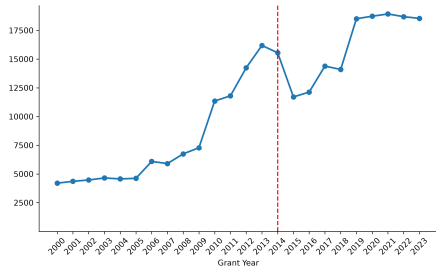
CPC Subclasses with High Alice Exposure

CPC Subclass	Short Description	Exposure (%)	
		OLS	GLMM
G06Q	ICT for administrative or business purposes	28.14	34.36
G07F	Coin or payment-operated apparatus	25.25	27.85
G16B	Bioinformatics and computational genomics	18.43	25.91
A63F	Games and amusement devices	20.51	24.81
G16C	Computational chemistry and molecular informatics	14.68	23.08
G07B	Ticketing, fare collection, and access control	18.07	18.06
G09B	Educational or demonstration apparatus	13.16	15.75
G16Z	ICT applied to specific fields	5.63	12.98
G01W	Meteorology-related measurement and forecasting	10.83	10.61
G07G	Registering cash registers, point-of-sale systems, and related devices	3.64	9.77
C12Q	Measuring or testing processes involving enzymes, nucleic acids, or microorganisms	4.87	6.28
G09C	Cryptography and secure communication	3.23	6.10
G01V	Geophysics; detection of underground structures or resources	4.87	5.82
G16H	Healthcare informatics	5.64	5.81
G01C	Measuring distances, navigation, and positioning	4.70	5.65
G10L	Speech analysis, recognition, and processing	4.50	5.42
G07C	Time or attendance registers; time recording devices	3.77	5.33

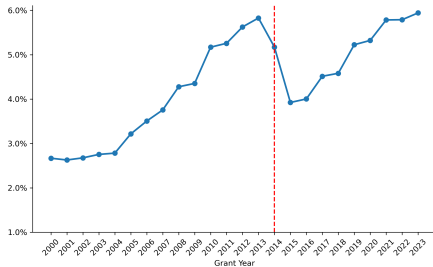
Weights between Primary and Secondary Classes

$N_{\text{subclasses}}$	$N_{\text{applications}}$ (Share)	OLS Estimates (%)		GLMM Weights (%)	
		Primary	Secondary	Primary	Secondary
1	313,143 (45.18)	118.91	–	100.00	0.00
2	222,836 (32.15)	65.93	24.67	70.28	29.72
3	102,798 (14.83)	53.61	31.07	54.18	45.82
4	38,470 (5.55)	43.12	40.22	44.08	55.92
5	12,883 (1.86)	29.12	58.63	37.16	62.84
6	4,336 (0.63)	22.22	88.63	32.11	67.89
7+	2,870 (0.41)	-3.49	51.33	28.27	71.73

Trend of Patent Grants in High-Exposure Technologies

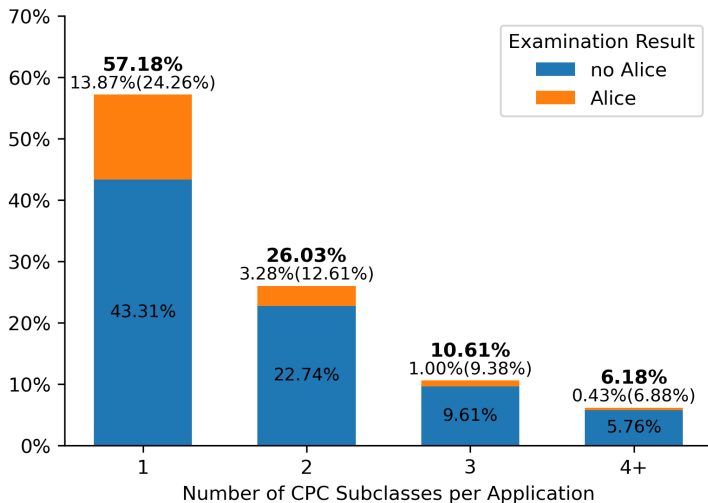


(a) Number of Patent Grants

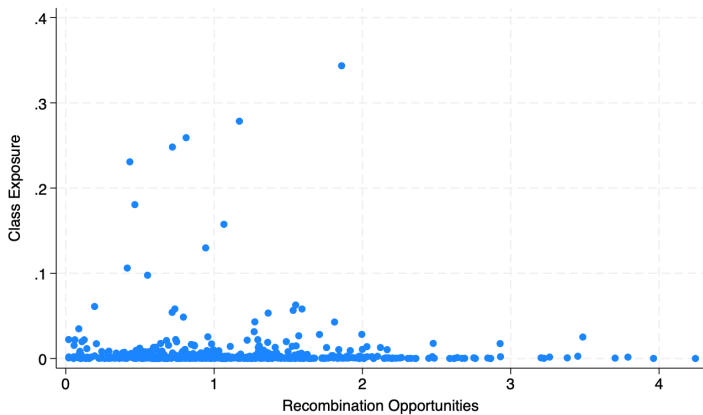


(b) Share of Patent Grants

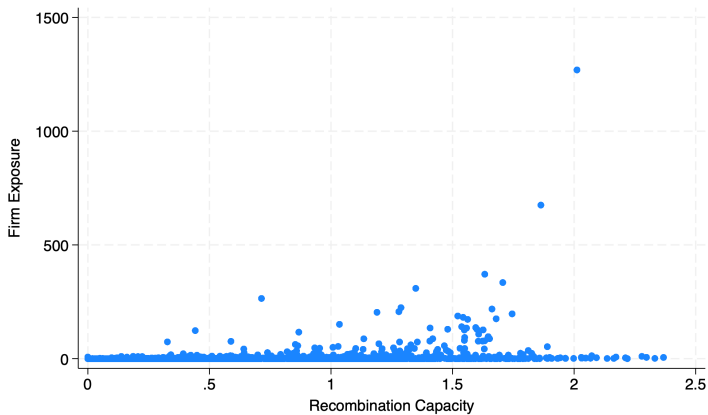
Cross-Class Recombination and Patent Exposure



Distribution of Recombination Opportunities



Distribution of Recombination Capacity



Changes of Inventor Teams

	Team Knowledge Breadth		Team Knowledge Dispersion		Member Knowledge Distance	
	(1)	(2)	(3)	(4)	(5)	(6)
Class Exposure $\times$ Post	0.189*	0.220**	0.0829**	0.0954***	0.0832***	0.0942***
	(1.88)	(2.24)	(2.32)	(5.91)	(2.59)	(6.78)
Recombination Capacity $\times$ Post		-0.000840		-0.00602***		0.000327
		(-0.07)		(-2.90)		(0.26)
Class Exposure $\times$ Recombination Capacity $\times$ Post		0.325		0.162*		0.123***
		(1.10)		(1.75)		(3.23)
Primary Class FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Primary Class	Primary Class	Primary Class	Primary Class	Primary Class	Primary Class
Pseudo/Adj. R^2	0.0760	0.0760	0.0364	0.0365	0.0587	0.0588
Num. of Observations	6586135	6586135	5801352	5801352	3760282	3760282
Model	PPML	PPML	OLS	OLS	OLS	OLS

Changes of Backward Citations

	Citation Breadth		Citation Dispersion	
	(1)	(2)	(3)	(4)
Class Exposure $\times$ Post	0.129 (1.63)	0.129** (2.57)	0.00331 (0.09)	0.0157 (0.97)
Recombination Opportunities $\times$ Post		-0.0359*** (-3.64)		-0.0172*** (-4.40)
Class Exposure $\times$ Recombination Opportunities $\times$ Post		0.403** (2.46)		0.217*** (3.23)
Primary Class FE	Yes	Yes	Yes	Yes
Filing Year FE	Yes	Yes	Yes	Yes
Clustering	Primary Class	Primary Class	Primary Class	Primary Class
Pseudo/Adj. R^2	0.0890	0.0892	0.1070	0.1077
Num. of Observations	4511111	4511111	4511111	4511111
Model	PPML	PPML	OLS	OLS

Changes of Backward Citations

Panel A: Comparison of Cited and Citing Patents

	Num. of Classes Increase	Num. of New Classes	I(Different Primary Classes)	Exposure Reduction
Class Exposure $\times$ Post	0.449*** (13.91)	0.316*** (12.20)	0.128*** (8.52)	0.110*** (53.84)
Cited Patent FE	Yes	Yes	Yes	Yes
Citation Lag FE	Yes	Yes	Yes	Yes
Clustering	Cited Patent	Cited Patent	Cited Patent	Cited Patent
Pseudo/Adj. R^2	0.4514	0.1929	0.1415	0.5748
Num. of Observations	21710221	21295412	20791074	21710221
Model	OLS	PPML	PPML	OLS

Panel B: Exposure Distribution of Backward Citations

	(1) Mean	(2) Median	(3) Mean of Top 25%	(4) Mean of Bottom 25%
Class Exposure $\times$ Post	-0.0413*** (-31.10)	-0.0490*** (-24.06)	-0.0314*** (-20.36)	-0.0395*** (-34.72)
Cited Patent FE	Yes	Yes	Yes	Yes
Citation Lag FE	Yes	Yes	Yes	Yes
Clustering	Cited Patent	Cited Patent	Cited Patent	Cited Patent
Pseudo/Adj. R^2	0.8380	0.7669	0.8071	0.7454
Num. of Observations	21710221	21710221	21710221	21710221
Model	OLS	OLS	OLS	OLS

Cited-Citing Patent Pairs

<i>Panel A: Comparison of Cited and Citing Patents</i>				
	Num. of Classes Increase	Num. of New Classes	I(Different Primary Classes)	Exposure Reduction
Class Exposure $\times$ Post	0.449*** (13.91)	0.316*** (12.20)	0.128*** (8.52)	0.110*** (53.84)
Cited Patent FE	Yes	Yes	Yes	Yes
Citation Lag FE	Yes	Yes	Yes	Yes
Clustering	Cited Patent	Cited Patent	Cited Patent	Cited Patent
Pseudo/Adj. R^2	0.4514	0.1929	0.1415	0.5748
Num. of Observations	21710221	21295412	20791074	21710221
Model	OLS	PPML	PPML	OLS
<i>Panel B: Exposure Distribution of Backward Citations</i>				
	(1) Mean	(2) Median	(3) Mean of Top 25%	(4) Mean of Bottom 25%
Class Exposure $\times$ Post	-0.0413*** (-31.10)	-0.0490*** (-24.06)	-0.0314*** (-20.36)	-0.0395*** (-34.72)
Cited Patent FE	Yes	Yes	Yes	Yes
Citation Lag FE	Yes	Yes	Yes	Yes
Clustering	Cited Patent	Cited Patent	Cited Patent	Cited Patent
Pseudo/Adj. R^2	0.8380	0.7669	0.8071	0.7454
Num. of Observations	21710221	21710221	21710221	21710221
Model	OLS	OLS	OLS	OLS

Continuation Children

<i>Panel A: Number of Continuation Children</i>				
	Continuation (rewrite claims)		Continuation-in-Part (add new matters)	
	(1)	(2)	(3)	(4)
	in 3 years	in 5 years	in 3 years	in 5 years
Post	0.0288 (1.47)	0.0445** (2.45)	-0.000256 (-0.01)	-0.00741 (-0.18)
Class Exposure × Post	0.345 (1.25)	-0.464* (-1.86)	0.733** (2.49)	0.588** (2.06)
Pending Duration	-0.730*** (-21.76)	-0.257*** (-20.05)	-0.104*** (-5.60)	-0.0605*** (-3.85)
Primary Class FE	Yes	Yes	Yes	Yes
Filing Year FE	Yes	Yes	Yes	Yes
Number of Obs.	1593744	1594940	1593012	1593793
<i>Panel B: Expand Number of Technology Classes by Continuity</i>				
	Continuation (rewrite claims)		Continuation-in-Part (add new matters)	
	(1)	(2)	(3)	(4)
	in 3 years	in 5 years	in 3 years	in 5 years
Post	-0.658*** (-4.22)	-0.820*** (-6.03)	-0.161*** (-3.30)	-0.220*** (-4.68)
Post × Class Exposure	0.00965 (0.01)	-1.218 (-1.30)	1.075*** (2.92)	0.898** (2.42)
Pending Duration	-0.154** (-2.50)	-0.0553 (-1.15)	-0.0429*** (-2.59)	-0.000319 (-0.02)
Primary Class FE	Yes	Yes	Yes	Yes
Filing Year FE	Yes	Yes	Yes	Yes
Number of Obs.	1289778	1385673	1582891	1585638

Changes of Firm Technological Boundaries

	Technology Scope		Existing Technology Reuse		New Technology Entry	
	(1)	(2)	(3)	(4)	(5)	(6)
Firm Exposure $\times$ Post	-0.00726** (-1.96)	-0.0869** (-2.49)	-0.0134*** (-3.03)	-0.109*** (-2.73)	-0.0181** (-2.23)	-0.101** (-2.11)
Recombination Capacity $\times$ Post		-0.0393* (-1.84)		-0.0689*** (-2.72)		-0.110*** (-5.32)
Firm Exposure $\times$ Recombination Capacity $\times$ Post		0.0301*** (2.67)		0.0368*** (2.88)		0.0349** (2.17)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm
Pseudo R^2	0.8772	0.8775	0.8831	0.8836	0.5557	0.5570
Num. of Observations	29998	29998	28511	28511	28775	28775